Doc. 309 – The Scale-Anchor Problem: ΛCDM and FFGFT Compared

Johann Pascher

July 2026

Contents

Doc. 309 – The Scale-Anchor Problem

What this is about

Both cosmological theories — ΛCDM and FFGFT — must introduce somewhere a number that couples the cosmic scale to the SI system. Neither derives this anchor from first principles.

This is not a coincidence — it is a structural problem of every physical theory that has dimensions. But the **kind and quality** of the anchor differ fundamentally. And there is an asymmetry that has not been explicitly named in any previous document: FFGFT inherits the systematics of the expansion reading through its scale fit.

This document is not part of the A-series. References: Doc. 190 (P20, P33), Doc. 267 (degeneracy), Doc. 279 (H_0 chain), Doc. 308 (cosmic sector).

Conventions

Legend.

[K]	*Core derivation* — derived mathematically or numerically from the foundational posits; reproducible.
[S]	*Structural claim* — justified but not yet fully proved.
[POSIT]	*Posit* — declared assumption, not further derived.

Central quantities.

$\xi = 4/30000$	Only fundamental parameter of FFGFT [POSIT, P33]. Form ($4/3$, 3, 2^2) forward-derived [K]; magnitude (5^4) intuitive/empirical [POSIT].
$\bar{\lambda}_e = \hbar/(m_e c)$	Reduced Compton wavelength of the electron; $= 3.8616 \times 10^{-13}$ m.
H_0	Cosmological parameter; in FFGFT: $H_0 = (\pi/2)\, c\, \xi^{10}/\bar{\lambda}_e = 66.82$ km/s/Mpc [K], exponent 10 [POSIT, P20].
$R_H = c/H_0$	Geometric maximal scale of the compact geometry — not an expansion horizon (Doc. 182).
ν_{Cs}	Caesium hyperfine transition; $= 9.192\,631\,770 \times 10^9$ Hz (exact, definition); conventional SI anchor.
$\nu_e = m_e c^2/h$	Electron frequency; $= 1.235 \times 10^{20}$ Hz; reading-independent reference frequency.
n	Integer exponent in $H_0 \propto \xi^n/\bar{\lambda}_e$; $n = 10$ is a posit (P20, Doc. 190).
Ω_Λ	Dark-energy density in ΛCDM; without reading-independent referent in the static reading.

Chapter A

The SI System — a Network without a First Anchor

A.1 All units traceable to one

Since 2019 all SI base units are defined via fixed natural constants. The seemingly "four independent measured values" c, h, e, k_B are physically not independent — they are all traceable to one single historically conventional anchor: the frequency of the caesium hyperfine transition ν_{Cs}.

Unit	Traceability	Historical origin
Metre	c/ν_{Cs} (numerical factor)	Earth meridian
Kilogram	$h\,\nu_{\mathrm{Cs}}^2/c^2$	Prototype kilogram
Ampere	$e\,\nu_{\mathrm{Cs}}$	Force definition
Kelvin	$h\,\nu_{\mathrm{Cs}}/k_B$	Thermometry

A.2 There is no first anchor

ν_{Cs} was chosen in 1967 because it was technically stable and realisable — not because it is physically distinguished. The Rb transition, the electron frequency, or a hundred other systems would have served equally. All present SI numerical values would then be different — all dimensionless ratios (m_e/m_p etc.) would remain identical.

What is truly fixed: only dimensionless ratios such as $m_e/m_p \approx 1/1836$. These are unit-independent.

Consequence: Every theory that predicts cosmological absolute values in SI implicitly states a relation between two frequencies — and must set some number to do so. The FFGFT statement

$$H_0 = \frac{\pi}{2}\,\frac{c\,\xi^{10}}{\bar{\lambda}_e}$$

is equivalent to the dimensionless statement:

$$\frac{H_0}{\nu_e} = \frac{\pi}{2}\,\xi^{10} = 2.79 \times 10^{-39} \qquad \text{with} \quad \nu_e = \frac{m_e c^2}{h}.$$

The SI reference is contained entirely in the choice of $\bar{\lambda}_e$ as the reference scale. Without that choice, H_0/ν_e is a pure number.

Chapter B

ΛCDM and its Scale Anchor

B.1 Six free parameters, all empirical

ΛCDM describes cosmology with six free parameters (Planck 2018): H_0, $\Omega_b h^2$, $\Omega_{\text{DM}} h^2$, n_s, σ_8, τ. All from observation — none derived from theory.

B.2 H_0: measured circularly

H_0 is determined via a four-rung distance ladder: parallax → Cepheids → SNe Ia → Hubble law. Each rung introduces its own calibration assumptions. The **Hubble tension** (Planck: 67.4, SH0ES: 73.0 km/s/Mpc, 5σ) is a direct consequence of this circularity.

B.3 The fine-tuning problem

Λ corresponds to a vacuum energy density $\rho_\Lambda \approx 5.96 \times 10^{-27}$ kg/m^3. QFT predicts $\rho_{\text{QFT}} \sim 5 \times 10^{96}$ kg/m^3. The ratio is 10^{123}. ΛCDM has no explanation — it sets Λ to the observed value without theoretical origin.

Chapter C

FFGFT and its Scale Anchor

C.1 Structure of ξ [K] and magnitude [POSIT]

$\xi = 4/30000 = 1/(3 \cdot 2^2 \cdot 5^4)$. According to P33 (Doc. 190):

- **Form** ($4/3$, factor 3, factor 2^2): forward-derived [K]
- **Magnitude** ($5^4 = 625$): intuitive/empirical [POSIT]

C.2 The actual fit: the exponent 10

ξ itself comes from the geometry. The **exponent 10** in $H_0 = (\pi/2)\, c\,\xi^{10}/\bar{\lambda}_e$ was chosen so that $H_0 \approx 67$ km/s/Mpc. Since $\xi \sim 1.3 \times 10^{-4}$, H_0 jumps between neighbouring exponents by a factor of ~7500:

n	H_0	Assessment
9	5.0×10^5 km/s/Mpc	unphysical
10	**66.8 km/s/Mpc**	plausible
11	8.9×10^{-3} km/s/Mpc	unphysical

There is no continuous fitting — $n = 10$ is the only integer choice that yields a physically sensible value. The justification remains: $n = 10$ gives a value compatible with ΛCDM measurements (P20, Doc. 190).

C.3 The inheritance: FFGFT inherits ΛCDM systematics [K]

67.4 km/s/Mpc is a ΛCDM measurement value — determined under the assumption of the expansion reading. Since $n = 10$ was calibrated to this value, FFGFT inherits the systematics of that reading.

The "natural" H_0 from the FFGFT redshift formula $z = e^{\xi x/\bar{\lambda}_e} - 1$ would be:

$$H_0^{\text{natural}} = \frac{c\xi}{\bar{\lambda}_e} \approx 3 \times 10^{33} \text{ km/s/Mpc} \qquad \text{(unphysical)}$$

To obtain ~67 km/s/Mpc, ξ^{10} is needed instead of ξ^1 — and the justification of this exponent is ultimately ΛCDM compatibility.

This is not a weakness of FFGFT alone — it is the unavoidable consequence of the degeneracy (Doc. 267): as long as both readings are empirically indistinguishable, every theory calibrates its parameters to the same data.

The remedy would be a reading-independent derivation of exponent 10 from the T^4 geometry forward — not yet achieved (P20, Doc. 190).

C.4 Cross-anchoring: what FFGFT achieves nonetheless

That ξ^{10} reproduces not only H_0 but also a_0 (inertia transition scale, Doc. 308) — that would not be expected from a pure H_0 fit. That is the cross-anchoring.

Sector	Quantity	ξ-dependence	Status
Cosmology	H_0	ξ^{10}	[K]
Inertia regime	a_0	ξ^{10}	[K]/[S]
Particle physics	lepton mass ladder	ξ-powers	[K]
Quantum mechanics	$K_{\text{frak}} = 1 - 100\xi$	ξ	[K]
Geometry	$k^* = \ln \varphi/\xi$	ξ	[S]

Ω_Λ in ΛCDM has **zero** such cross-anchorings.

Chapter D

Comparison

Property	ΛCDM	FFGFT
Free scale anchors	6–7, empirical	1 (ξ, P33)
Theoretical origin	none	none
Cross-anchoring	no	yes (5+ sectors)
Circularity	yes (H_0 ladder)	no
Inheritance of ΛCDM systematics	inherent	via exponent 10
Fine-tuning	10^{123} (Λ)	absent structurally
Hubble tension	5σ, open	structurally absent

FFGFT exchanges six isolated parameters for one cross-anchored — no proof, but a reduction with direction. The inheritance of the ΛCDM scale value via exponent 10 remains the central open weakness.

Chapter E

Open Points (Doc. 190)

- **P20:** Form of H_0 as ξ-ratio fixed; exponent 10 is external calibration from ΛCDM, not derived.
- **P33:** Magnitude of ξ (factor 5^4) intuitive/ empirical, no forward proof. Structural parallel to P20.
- **Inheritance (new, Doc. 309):** Exponent 10 calibrated to $H_0^{\Lambda\text{CDM}}$. Remedy: reading-independent derivation from T^4 geometry — not yet achieved.